

KEDAR DAMALE

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Portfolio · GitHub · LinkedIn

SUMMARY

Applied Data Scientist with 1.5 YOE building end-to-end AI/ML solutions across fintech, financial automation, pharma, and medical research. Skilled in Data Science, Machine Learning, Generative AI, multi-agent systems, model development, data pipelines, and production deployment, with experience taking solutions from research and experimentation to production-ready systems.

SKILLS

- **Foundational Mathematics** : Linear Algebra, Calculus, Probability and Statistics
- **Programming and Data** : Python, NumPy, Pandas, Polars, SQL
- **Generative AI** : LangChain, LangGraph, Multi-Agent Systems, RAG, HITL, MCP
- **Machine Learning** : Scikit-learn, XGBoost, Feature Engineering, Model Evaluation, Hyperparameter Tuning
- **Deep Learning** : PyTorch, CNNs, YOLOv8, Transfer Learning, Model Training and Evaluation
- **DevOps** : FastAPI, Docker, AWS, MLflow, Git/GitHub, CI/CD, ETL/ELT Pipelines

WORK EXPERIENCE

ML and GenAI Engineer

Globalspace Technologies Ltd.

Budget Automation and Incentive Calculator

Jan 2026 – Present

Navi Mumbai, MH, India

- Replaced an XGBoost model per employee—which became difficult to manage beyond 20–30 people—with a unified hierarchy model designed to scale to thousands of medical representatives (MRs). The solution is under review by large pharmaceutical companies, including Himalaya and Zydus.
- Built a top-down budgeting workflow that sets targets using prior performance and capacity, supports time-bound revision requests, reconciles conflicting requests where possible, and routes proposed changes to managers for approval.
- Developed a DAG-based incentive calculator with a custom AI agent that converts policy PDFs into executable functions at runtime and explains each employee's result, enabling business users to update policies without specialist admin support.

PatGPT

- The earlier workflow relied entirely on retrieval-augmented generation (RAG). It handled simple lookups but could not answer multi-step commercial questions or recommend actions across more than 3 million pharmaceutical sales and tender records.
- Rebuilt PatGPT as an agentic planning assistant for forecasting, simulation, prediction, and historical analysis, with human-review checkpoints and evidence-grounded answers.
- Added streamed responses, safe retry handling, and checksum-verified data promotion and rollback. The system is now used at Zydus.

Freelance Automation Developer

Propelligence Advisors Pvt. Ltd.

May 2025 – Jan 2026

Thane, MH, India

- Manual reconciliation of purchase-register invoices against GSTR-2B records took five to six days per cycle; inconsistent vendor, invoice, and date formats made exact-match tools unreliable.
- Engineered a large-file reconciliation pipeline using RapidFuzz with an 80%+ match threshold and date/value tolerances, Polars processing, and constant-memory report generation.
- Processed 10,000+ invoices in 12 seconds, down from five to six days per reconciliation cycle—over 99.9% less time—with 100% precision (no false positives). Integrated the module into Granska, an audit suite adopted by two CA firms and about 20 auditors.

PROJECTS

1. Chessablanka – Chess Position Analyzer | GitHub

Apr 2025 – Jan 2026

- Converting a chessboard photo into a position an engine can analyze usually requires manual transcription, making the process slow and error-prone.
- Built an end-to-end computer-vision pipeline with OpenCV to detect the board and map its grid. Trained a custom YOLOv8 model on a Roboflow-augmented dataset expanded from 700+ to 1,700 images.
- Achieved 98.57% detection accuracy on 500+ test images and integrated Stockfish for immediate position evaluation and best-move analysis.

2. NeuroTRIBE – Cortical Response Analysis for ADHD Research | GitHub

Aug 2026 – Present

- A reproducible analysis was needed to test whether ADHD is associated with differences in stimulus-evoked cortical responses.
- Built a TRIBE v2 analysis pipeline using Healthy Brain Network movie-fMRI data to compare participants with confirmed ADHD against a matched cohort.
- Added Docker Compose orchestration for repeatable cohort-level exploration and kept legally restricted inputs outside the workflow so their access requirements do not block setup.

3. Cattle Monitoring Using DBSCAN and ESP32 | GitHub

Oct 2025 – Nov 2025

- Farmers monitoring free-grazing cattle relied on manual rounds, making it difficult to track herd locations and recurring grazing patterns.
- Led a team that built a distributed tracking system: one ESP32 master and multiple child nodes sent GPS and RSSI readings to a Flask and MongoDB backend.
- Applied DBSCAN to identify grazing zones and mapped them on a live Next.js dashboard, giving farmers a clearer view of herd movement.

EDUCATION AND ACHIEVEMENTS

- **Qualified GATE** — Data Science and Artificial Intelligence 2026
- **Graduated in IT engineering with 8.5 CGPA from Mumbai University** —
- **Published Automation Suite for CA & Audit Firms** — Adopted by 2 CA firms and currently being used by about 20 auditors